Phase 0 Deep Reconnaissance: Elastane × AI-Molecular-Design — Technical Intelligence Map

TL;DR

- **Elastane (segmented polyurethane-urea) is the single most consequential blocker to fibre-to-fibre recycling of children's/baby stretch apparel, and the genuinely unsolved seam at the AI × chemistry intersection is the *de novo computational design of a urethanase enzyme or organocatalyst for selective carbamate cleavage in elastane fibres* — every demonstrated breakthrough on PU/elastane to date (GRASE/AbPURase, UMG-SP variants, FAST-PETase-style work) uses ML to mine or engineer natural enzymes, never to design one from scratch.**
- **The most credible pure-software contribution with maximum ceiling is a publicly released "designed-for-elastane" theozyme placed into a de novo scaffold via RFdiffusion2/RFdiffusion3 + LigandMPNN + PLACER**, since (a) Bauer/Baker (bioRxiv Nov 2025) just demonstrated de novo amide-bond cleavage at $>10^8$-fold acceleration with a 36% one-shot hit rate, (b) Lauko et al. (*Science* 388:eadu2454, 2025) demonstrated de novo serine hydrolases with kcat/Km up to $2.2\times10^5\ M^{-1}\ s^{-1}$, and (c) no group has yet combined these tools on the Ser-Sercis-Lys amidase-signature triad that natural urethanases use. ADS
- **Adjacent flag-planting territories** include: ML-screened selective solvent systems (extending the DMSO+DBN Ferreira et al. 2025 paradigm via COSMO-RS + GNNs); ML-discovery of bimetallic or organocatalysts for selective urethane cleavage analogous to the Byron/Sadow/Delferro 2026 PE-hydrogenolysis work; and digital-twin/sortation pipelines fusing hyperspectral fiber detection (Refiberd, Specim) with downstream chemistry recommendation.

PART A — The Elastane Problem in Full Technical Detail

A1. Elastane chemistry and why it resists degradation

Elastane (spandex/Lycra) is a **segmented thermoplastic polyurethane-urea block copolymer** with two distinct phases:

- **Soft segments (~65–85 wt%)**: polyether macrodiols, most commonly **poly(tetramethylene ether glycol)** (**PTMG**, also called PTMEG), Mn typically 1000–2000 Da; in some grades polyester polyols are used. The Lycra Company publicly confirms PTMEG is ~70% of fibre content and MDI is 15–20% (Lampoon Magazine, citing The Lycra Company). Lampoon Magazine

- **Hard segments (~15–35 wt%)**: aromatic **4,4'-methylene diphenyl diisocyanate (MDI)** extended with short diamines (typically ethylenediamine), giving **urea linkages** in addition to urethane bonds. The Lycra Company has now partnered with Qore/Cargill to commercialize **QIRA**, a bio-derived **1,4-butanediol** feeding into bio-PTMEG and enabling "70% renewable content" Lycra (lycra.com press release). LYCRA

The hard segments **microphase-separate** from the soft segments, forming **hydrogen-bonded urea aggregates** that act as physical (not covalent) crosslinks. The soft-segment Tg is well below room temperature (~ −60 to −80 °C for PTMG), while hard-segment domains have effective service temperatures into the 175–200 °C range; **above ~230–250 °C the urea/urethane begins to decompose oxidatively rather than melt cleanly**. TGA on elastane (Ferreira et al. 2025, *Polymers* 17:3247) shows three decomposition stages: initial event around **257 °C**, hard-segment degradation steps in the 280–330 °C window, and a third event near **418 °C** for the soft segment. Because PET fibre melt-spinning sits at **265–290 °C**, elastane is *guaranteed* to be at the onset of thermal degradation in any PET-melt recycling stream. ResearchGate + 2

A2. Why 2–5% elastane defeats every recycling stream

- **Mechanical shredding**: Elastane filaments retain elasticity through the shredder, causing **clumping, filter-clogging, and machine soiling**. Boschmeier (TU Wien, quoted in *International Fiber Journal*): "elastane leads to soiling, blockages and clumping in the machines." Reduces blade life and chokes fibre opening lines. MDPI Fiberjournal
- **Thermomechanical / melt extrusion**: Elastane decomposes at PET melt temperatures, releasing **MDA (4,4'-methylene dianiline, a Cat-2 carcinogen)**, CO_2, and chain-extender fragments; nitrogen-containing degradation products contaminate the PET melt and reduce intrinsic viscosity / discolour the pellet.
- **PET glycolysis / methanolysis / hydrolysis**: Elastane is *not* inert. Yan et al. (*Science Advances* 2024, DOI 10.1126/sciadv.ado6827) demonstrated that under microwave-assisted **ZnO-catalysed glycolysis at 210 °C, "polyester was completely converted to BHET, and spandex was transformed into diphenylmethane-containing molecules and sticky polyols"** — elastane consumes ethylene glycol, dilutes BHET purity with MDA-derived contaminants, and produces tacky polyol residues that foul the reactor. Conversely, in **alkaline hydrolysis**, elastane *also* depolymerises (PMC12944338), so it cannot be reliably "left behind" and contaminates the terephthalic acid product. THF-co-solvent neutral hydrolysis preserves spandex only at the cost of accepting a contaminated polyester residue. Science ScienceDirect
- **Enzymatic depolymerisation (e.g., LCC-ICCG, FAST-PETase, Carbios)**: Elastane physically wraps and "encapsulates" PET microfibres in the core-spun yarn, **sterically blocking enzyme access** to the PET surface; the MDI-aromatic surface is not a substrate for PET hydrolases.
- **Pyrolysis**: Elastane introduces **organic nitrogen** into the pyrolysis oil (MDA, anilines, isocyanate fragments), which poisons downstream catalysts and is a regulatory problem for

fuel/feedstock use.

A3. Why this is acute for children's/baby apparel

Pre-/post-consumer audits and product specs consistently show elastane in 2–8% bands in: **onesies and bodysuits (cotton/elastane 95/5 or 92/8)**, **leggings, jeggings, and tights (cotton/poly/elastane 70/25/5 to 60/35/5)**, **socks, sleep-and-play, and rib waistbands**, and increasingly in **swaddles and stretch-knit baby blankets**. Per Ferreira et al. (*Polymers* 17:3247, 2025): elastane "represents about 1.1% of the global fibre market … but is incorporated in a very large fraction of apparel products, typically in the range of 2–10 wt% for stretch fabrics, although higher contents may be used in specialised compression or shapewear." Children's stretch garments cluster at the high end of this band.

The **dermal/infant-safety overlay** is acute: (a) elastane that has thermally aged on prolonged hot-wash cycles can release MDA (regulated under EU REACH Annex XVII as a SVHC), and any recycled feedstock with residual MDA carries cross-contamination risk for baby skin; (b) regulators (EU CPSR, US CPSIA) demand higher purity for infant garments than for adult apparel; (c) the OEKO-TEX Standard 100 Class I (textiles for babies) sets the tightest aromatic-amine limits.

PART B — State of the Art in Elastane Separation, Degradation, and Recycling

B1. Selective dissolution / solvent approaches

- **DMSO + DBN (Ferreira et al., *Polymers* 17:3247, 2025, DOI 10.3390/polym17243247)** is the current benchmark for *selective elastane destruction*. Verbatim: **"DMSO alone promoted only partial elastane mass loss, typically 26–32% at 80–100 °C (60 min) and up to 79% at 120 °C (10 min), whereas the addition of 0.1% v/v DBN enabled near-complete or complete mass loss (81–100%) across 80–120 °C within 10–60 min. Complete removal of elastane was achieved in isolated elastane filaments at 100 °C within 30–60 min."** On real mixed waste (94/6 cotton/elastane and 87/13 PET/elastane) treatments at 100 °C/30–60 min produced 9–14% and 7–13% area increase (i.e., loss of elastic recovery) while **largely preserving cotton and PET tensile strength**. DBN (1,5-diazabicyclo[4.3.0]non-5-ene) acts as a strong organic base catalyst promoting hydroxide/DMSO-mediated cleavage. This is the *current* highest-yield, lowest-temperature, most-selective system, and explicitly the launchpad seam for ML solvent design. nih nih
- **Older / adjacent solvents**: DMF (Boschmeier et al., 70 °C, 50:1 solvent-to-fiber, removes elastane by *dissolution* not degradation), DMAc, NMP, and ionic liquids. All trade off energy use, solvent recovery, and toxicity. ResearchGate
- **THF co-solvent neutral hydrolysis** preserves spandex while degrading dyed PET; orthogonal route that produces a polyester-free spandex residue still to be handled.

ScienceDirect

B2. Chemical depolymerisation of PU / elastane

- **ZnO-catalysed microwave glycolysis** (Yan et al., *Science Advances* 2024, DOI 10.1126/sciadv.ado6827): 210 °C, 15 min, MDI hard segments → "diphenylmethane-containing molecules + sticky polyols" while PET → BHET, cotton/nylon intact. One of the only published routes that *co-depolymerises* elastane and PET, but recovered MDI-related fraction is impure. Science
- **Solid-state alkaline depolymerisation** (PMC12944338): 95/5 and 85/15 PET/elastane blends, NaOH 2.1 mol/PET repeat unit, 140 °C, 5 min in a lab kneader; quantifies the elastane parasitic base consumption and the contamination of the TPA product. PubMed Central
- **Solvolysis/glycolysis of PU foams** (PMC9319047): well-established at industrial scale for rigid PU; recovered polyol quality is the limiting factor; chemistry has not been cleanly transferred to elastane *fibres* because fiber morphology (highly crystalline, dyed, dense) is unlike open-cell foam. nih
- **GRASE-AbPURase chemo-enzymatic glycolysis** (Chen et al., *Science* 390:503–509, 30 Oct 2025, DOI 10.1126/science.adw4487) is the new state of the art: **AbPURase (a metagenomic urethanase discovered by a GNN) shows two orders of magnitude greater activity than known enzymes in 6 M diethylene glycol, achieving near-complete (98.6%) depolymerisation of kilogram-scale commercial PU within 8 hours**. This was on PU foams, not elastane fibre — the elastane fibre transfer is the obvious next step and not yet published. PubMed + 2

B3. Enzymatic degradation of PU / elastane

- **UMG-SP1/2/3 family** (metagenomic urethanases, amidase-signature superfamily, **Ser-Sercis-Lys triad**). Structure-guided engineering (Li et al., *Adv. Sci.* 12:2416019, 2025, DOI 10.1002/advs.202416019): single mutants A141G, D226A, Q399A; A141G gives **>30.7-fold improved degradation of polyester-PUR particles**, Q399A releases 7.4× more MDA from MDI-DEG. Wiley Online Library
- **Aes72** (rational mutant P367A/L381A, *ACS Catalysis* 2025, DOI 10.1021/acscatal.5c01062): 2.8-fold WT activity on BTC, 10.9× WT on commercial polyester-PU foam. ACS Publications
- **Three new drain-metagenome urethanases on MDA-based elastane model substrates** (*Green Chemistry* 2025, DOI 10.1039/D5GC03560K) — the first published demonstration of urethanases on elastane-derived substrates; all sequence-mining, no design.
- **Mechanism (critical for AI design)**: Paiva et al., *Chem. Sci.* 16:2437–2452 (2025), DOI 10.1039/D4SC06688J. Verbatim: "These enzymes are capable of hydrolyzing urethane bonds … employing a Ser-Sercis-Lys triad for catalysis, similar to members of the amidase signature protein superfamily … STAGE 1 comprises the rate-limiting step … cleavage of

the substrate's urethane bond by its ester moiety and the release of the alcohol-leaving group (overall Gibbs activation energy of 20.8 kcal·mol^{-1})." QM/MM follow-up (Fernandes/Ramos, *JACS* 147:42511–42523, 2025, DOI 10.1021/jacs.5c13147) confirms an "esterase-like three-step mechanism (acylation, hydrolysis, decarboxylation)" requiring alkaline media for Lys deprotonation. **Crucially, this is *not* the canonical Ser-His-Asp triad of PETases/cutinases**, which is why FAST-PETase-style transfer is non-trivial. PubMed + 2

B4. Engineered / dissolvable elastane and design-side approaches

- **Chitosan finishing for selective recovery** (Grosvenor, Cohen, Chazot et al., *RSC Applied Polymers* 3:1193–1203, 2025, DOI 10.1039/D5LP00213C): a 4 wt% chitosan-in-0.5 N HCl dip yields a **5–10 µm coating** that preserves elastic modulus and energy dissipation under cyclic strain, but the coating can be **redissolved under mild acidic conditions**, releasing the elastane filament from core-spun yarns. Data openly archived (Zenodo DOI 10.5281/zenodo.15863582). This is a design-for-disassembly route that ML could optimise. RSC Publishing RSC Publishing
- **Roica V550 (Asahi Kasei)**: world's first Cradle-to-Cradle-Gold-certified stretch yarn, partial degradation under ISO 14855-1 tested by OWS, with Asahi Kasei stating it decomposes to CO_2 and water; per The Slow Label citing Asahi Kasei/OWS test data, **"35% of the ROICA™ V550 yarn breaks down within 270 days"** (not "fully biodegradable" by regulatory definition). Asahi Kasei + 2
- **LYCRA T400 (elastomultiester / elasterell-p)**: 40% 3GT + 60% 2GT bicomponent PET-only fiber producing stretch via differential shrinkage crimping. The Lycra Company has demonstrated in glycolysis pilots that T400 "does not disrupt standard polyester recycling streams." Fiberjournal Fiberjournal
- **NEOLAST (Celanese × Under Armour, Jan 2024)**: recyclable elastoester polymer; solvent-free melt-extrusion; explicit positioning as elastane-replacement enabling future polyester-stream recyclability. Plastics Engineering
- **Roica EF**: GRS-certified recycled elastane with 58% production-waste content. Première Vision
- **Renewable Lycra with QIRA (2024 onward)**: bio-derived BDO → PTMEG; same fibre chemistry, lower carbon footprint, no end-of-life benefit per se.
- **Resortecs** (Smart Stitch + Smart Disassembly): heat-dissolvable threads at 150/170/200 °C and thermal-disassembly ovens; per Resortecs' own technology page, **"Smart Stitch™ & Smart Disassembly™ ensure a 90% recyclability rate,"** with another Resortecs product page citing "recyclability rates as high as 95%." Removes elastic trims as a *complement* to chemistry-based elastane handling. Ellen MacArthur Foundatio...

B5. Industrial / startup activity specifically targeting elastane

- **Samsara Eco** (Australian, founded 2021, ex-ANU). EosEco platform; biophysics +

chemistry + biology + computer-science/AI ("library of plastic-eating enzymes"); claims **20-minute** enzymatic depolymerisation; commercial Jerrabomberra (NSW) plant opened 3 Sept 2025; Nilit Southeast Asia nylon 6,6 plant due 2026. Per the joint Lululemon/Samsara press release (June 12, 2025), **"This multi-year agreement could see Samsara Eco's materials support approximately 20% of lululemon's overall fibers portfolio and advance its progress towards making more products with preferred materials by 2030."** The language is conditional ("could see"), the near-term milestone year is **2030** (lululemon's preferred-materials goal), and the fiber types specified are **recycled nylon 6,6 and polyester — not elastane. Critically: Samsara explicitly says its process "can handle ... nylon/elastane blends," but published claims describe enzymatic depolymerisation of the *PA* component (PET, nylon 6, nylon 6,6); elastane appears as un-converted residue, not as a depolymerised stream**. PU/elastane is a *future* claim, not deployed. Textileworld + 2

- **Carbios** (France): C-ZYME (LCC-ICCG) enzyme; first bio-recycled white polyester T-shirt Oct 2024. Per Carbios FY2023 results (GlobeNewswire, April 2024) and BNP Paribas, Carbios "has successfully raised more than **€250 million** in equity through two record-breaking capital increase transactions, first in 2021, and again in 2023," with the July 2023 raise being **€141 million gross — "the largest fundraising on Euronext Growth since 2015."** Nylon and PA6 expansion announced; no published PU/elastane process. BNP Paribas Carbios
- **Worn Again Technologies** (Switzerland): multi-solvent chemical recycling of polycotton; the Winterthur Accelerator plant (March 2026) demonstrates **>95% solvent recovery** with **explicit elastane and acrylic removal as a "novel separation process"** — the company treats elastane as a separable contaminant, not as a valorised stream. Worn Again Worn Again
- **CuRe Technology** (Netherlands): glycol-based polyester rejuvenation; not elastane-specific but PU/elastane tolerance is part of its pitch.
- **Sortile** (US): AI-driven sortation upstream of chemical recycling.
- **Refiberd** (US, Bajaj/Gupta/Wang, founded May 2020): hyperspectral imaging + ML; explicitly claims **detection of trace spandex/nylon/acrylic below 2% composition** at **millisecond per garment** speeds; "one camera unit will eventually be able to sort up to 7 million pounds of material annually" (Sourcing Journal/WWD). Refiberd
- **Specim RETEX / FX17** and **PICVISA**: industrial hyperspectral textile sortation; both companies explicitly call out elastane identification as one of the toughest open problems ("tackling the identification of elastane has proven challenging" — Specim case study with PICVISA). Specim
- **The Lycra Company**: pilot-scale recycling of spandex fibres from garments (per CEO Gary Smith, *WWD* 2024); QIRA bio-derived line at commercial scale 2024+; no published depolymerisation chemistry. WWD
- **Hyosung Creora Regen**: recycled spandex from pre-consumer waste; modest scale.

PART C — The AI/ML for Molecular & Catalyst & Enzyme Inverse Design Attack Vector

C1. The Byron/Sadow/Delferro/Ferrandon 2026 ACS Catalysis paper

Byron, McCullough, Hall, Ammal, Heyden, Bryant, Wang, Tonk, Wen, Kropf, Huang, Sadow, Delferro, Ferrandon. "Data-Driven Discovery of Bimetallic Nanoparticles Catalysts for the Hydrogenolysis of Polyethylene." ***ACS Catalysis*** **16(1):431–445 (1 Jan 2026). DOI: 10.1021/acscatal.5c06514.** ACS Publications

This is the most current high-impact reference paper for ML-driven catalyst discovery for polymer deconstruction. The Argonne/Ames/USC team (Delferro, Sadow, Heyden, Ferrandon are the leaders of the U.S. DOE iCOUP institute, whose Pt/$SrTiO_3$ PE-hydrogenolysis line dates to 2019; their bimetallic Ru-M_3 paper, *Chem Bio Eng* 2023, DOI 10.1021/cbe.3c00007, established product-distribution tunability) used a data-driven workflow over candidate Ru-, Pt-, Ni-, Co-, Fe-based bimetallic compositions, combining DFT-calculated descriptors, experimental rate/selectivity data from prior team work, and ML regression to recommend composition–support pairs. Its transferable methodology comprises: (a) curated descriptor set including d-band center, segregation energy, ensemble effects; (b) Bayesian or kernel regression over a small experimental dataset; (c) wet-lab validation feedback loop. **Direct transfer to PU/elastane is non-trivial** because (i) C–C hydrogenolysis breaks σ-bonds where PE has no functional groups, while elastane requires *selective* cleavage of urethane C–N or C–O bonds *in preference to* other C–C and C–N bonds, (ii) elastane is heterogeneous (soft polyether + hard urea) so hydrogenolysis would over-cleave, and (iii) the relevant catalysis is not high-pressure H_2 but rather low-temperature solvent-assisted bond cleavage. Nonetheless, the *methodological scaffold* (small descriptor set + ML + tight experimental loop) is directly portable to a catalyst-screening study for urethane cleavage.

C2. The broader landscape of AI/ML for catalysis

- **Machine-learned interatomic potentials (MLIPs)**: MACE (Batatia et al.), NequIP/Allegro (Batzner, Smidt), CHGNet (Deng et al.), SevenNet, GemNet-OC, EquiformerV2 — the universal-MLIP family that emerged 2022–2025 and now achieves near-DFT accuracy at orders-of-magnitude lower cost. Wikipedia
- **Open Catalyst Project** (Meta FAIR + CMU): **OC20** (1,281,040 DFT relaxations / >260M single-point calculations / 82 adsorbates × 11,451 metallic slabs / 55 elements) and **OC22** (oxides, ~62,000 relaxations with spin-polarisation + Hubbard-U). Both datasets are open under CC-BY-4.0; models and code on the fair-chem GitHub. OC25 has extended to solid-liquid interfaces. PubMed Central Emergent Mind
- **Other dataset/model families**: MatPES (matpes.ai), Materials Project, **GNoME (Google DeepMind, Merchant et al.,** ***Nature*** **623:669–674, 2023, DOI 10.1038/s41586-023-06735-9)** — predicted 2.2 million new crystal structures, **"of which 380,000 are the most stable,**

making them promising candidates for experimental synthesis" (45× larger than the number of such substances identified in the entire history of science to that point). Other resources: Alexandria, MPtrj, OMat24, ODAC23 (MOFs). Google DeepMind Biocomm

- **Generative chemistry/material models**: MatterGen (Microsoft Research, generative crystal design), CrysFormer, Open Reaction Database for retrosynthesis training.
- **Bayesian/active-learning for reaction optimisation**: Atinary, Summit, EDBO (the Doyle/Dreher Bayesian optimisation work in coupling catalysis).
- **All tools are open-source and free for academic use**; FAIR-Chem and MACE are the most plug-and-play.

C3. AI for enzyme engineering applicable to PU/elastane

- **Protein structure prediction**: AlphaFold2/3 (DeepMind), RoseTTAFold (Baker), ESMFold/ESM-2/ESM-3 (Meta FAIR / EvolutionaryScale), OmegaFold.
- **Generative protein design**:
    - **ProteinMPNN** (Dauparas et al., 2022, Baker lab): inverse-folding sequence generation onto a target backbone.
    - **RFdiffusion** (Watson et al., *Nature* 2023): diffusion model generating de novo backbones, with active-site scaffolding for enzyme design. Baker Lab
    - **RFdiffusion2 / RFdiffusion3** (Ahern et al., *Nat. Methods* 23:96–105, 2026, DOI 10.1038/s41592-025-02975-x; Butcher/Krishna/Baker RFdiffusion3 bioRxiv 10.1101/2025.09.18.676967, Nov 2025): atom-level active-site conditioning for serine hydrolases, retroaldolases, cysteine hydrolases, zinc hydrolases, DNA-binding proteins.
    - **LigandMPNN**: substrate-conditioned sequence design.
    - **Chroma** (Generate Biomedicines), Genie 2, **salad** (*Nat. Mach. Intell.* 2025), Proteus2.
- **The FAST-PETase precedent** (Lu, Diaz, Alper et al., *Nature* 604:662–667, 2022, DOI 10.1038/s41586-022-04599-z): UT Austin used the **MutCompute** 3D self-supervised CNN to predict mutations that stabilise PETase folds. Five mutations N233K/R224Q/S121E/D186H/R280A yielded **up to 38× higher activity at 40–50 °C** and degraded 51 different post-consumer PET products within hours-to-weeks. Canonical proof that ML-aided rational mutation can drop the activation barrier and tolerate substrate heterogeneity for polymer-degrading enzymes. Nature
- **The new state-of-the-art for PU specifically — GRASE**: Chen et al. *Science* 390:503–509 (2025), GNN-based recommendation framework using protein sequences + predicted apo structures + N-aryl-carbamate substrate descriptors. **Pythia-Pocket** for structure-conditioned features + a supervised activity predictor + a stability classifier. **AbPURase** hit (from *Acinetobacter baumannii*): two orders of magnitude more active than known urethanases in 6 M DEG, **98.6% PU foam depolymerisation in 8 h at kilogram scale**. PubMed + 3

- **De novo design of carbamate-cleaving enzymes — the open frontier**: As of June 2026, **no published paper or preprint applies generative protein design (RFdiffusion family, ProteinMPNN, ESM-IF, ESM3, LigandMPNN) to design a urethanase from scratch**. The nearest analogues are: (a) **Lauko, Pellock, Sumida et al., "Computational design of serine hydrolases," *Science* 388:eadu2454 (2025), DOI 10.1126/science.adu2454** — kcat/Km up to 2.2×10^5 M^{-1} s^{-1}, on activated esters with the classical Ser-His-Asp triad; (b) **Bauer/Baker et al., "Computational design of metalloproteases," bioRxiv 10.1101/2025.11.20.689622 (Nov 2025)** — verbatim: "In a single one-shot design round of 135 designs, 36% of the designs had activity … the most active design accelerated peptide bond hydrolysis more than 10^8-fold over the uncatalyzed reaction" — but on peptide amide bonds, not aryl carbamates; (c) Kim, Woodbury, Ahern et al., "Computational design of metallohydrolases," *Nature* 649:246–253 (3 Dec 2025), DOI 10.1038/s41586-025-09746-w — kcat/KM up to 53,000 M^{-1} s^{-1} on activated esters. The required theozyme for elastane is a **Ser-Sercis-Lys** constellation (per Paiva *Chem. Sci.* 2025 mechanism) with a planar carbamate oxyanion hole and a deep aromatic S1 pocket to accommodate MDI/MDA. **This has not been designed.** bioRxiv Nature

C4. AI for solvent / reaction discovery

- **COSMO-RS** (Klamt) for predictive solubility/Gibbs-energy of mixing in any solvent–solute pair; now extensively coupled to ML.
- **Aoki et al.** (*Macromolecules* 2023, DOI 10.1021/acs.macromol.2c02600) — multitask neural network on Flory–Huggins χ data for polymer-solvent miscibility prediction.
- **Qu et al., *J. Cheminformatics* 17 (2025), DOI 10.1186/s13321-025-01018-z** — ML-driven generation and screening of ionic liquids for cellulose dissolution.
- **Selective dissolution of PVC** in mixed plastic waste via COSMO-RS-screened solvents (ResearchGate 388567180) demonstrates the workflow applicable to elastane: oligomer MD → COSMO surface → temperature-dependent χ for elastane in candidate solvents and base catalysts.
- **Nurasyqin et al., *Catalysts* 15:1154 (2025)** uses COSMO-RS + experiment to screen deep eutectic solvents for PET hydrolysis. **No equivalent study has yet been published for selective elastane dissolution/cleavage** — a true open seam.

C5. Datasets and benchmarks

- **Polymer**: PoLyInfo (NIMS, half-million data points, manually curated for two decades), PI1M (~1M generative polymer SMILES), PolyMetriX, the Cambridge Polymer Genome (CPG, Ramprasad lab).
- **Reaction / catalysis**: Open Reaction Database (ORD), USPTO reaction corpus, Reaxys (commercial), the iCOUP / Argonne plastic-deconstruction datasets (Delferro group).

- **Protein**: UniProt, BFD, ProteinNet, OAS for antibodies, PDB; **RemeDB** (Pollutant Degrading Enzymes from metagenomic sequences) is the only enzyme database explicitly labelled for plastic substrates (bioRxiv 2025.02.09.637306). biorxiv
- **Textile-recycling-specific datasets: the major chokepoint.** No openly published hyperspectral textile dataset of elastane-blend NIR spectra at industrial throughput exists; Refiberd, Specim, Tomra hold this data privately.

C6. Honest assessment: what AI can credibly contribute on elastane

Demonstrable as a pure-software result (no wet lab required):

1. A **de novo urethanase theozyme** for the Ser-Sercis-Lys triad placed in an RFdiffusion2/3-generated scaffold, with LigandMPNN sequence assignment, PLACER ensemble scoring against a carbamate transition state, and AlphaFold3/ESMFold self-consistency validation — **a publishable in-silico result** equivalent to Lauko et al. up to the wet-lab step.
2. An **ML-screened selective solvent/catalyst library** for elastane disruption using COSMO-RS-on-oligomer + multitask GNN, benchmarked against the DMSO+DBN Ferreira gold standard, with concretely predictable mass-loss/selectivity targets.
3. A **digital-twin pipeline** combining publicly available hyperspectral fibre spectra + downstream chemistry recommendation, plus uncertainty-aware "is this garment recyclable in stream X?" classifiers for sortation.
4. **Mechanism-guided fine-tuning of a universal MLIP (MACE-MP-0)** on Ru/Pt/Mo bimetallic carbamate cleavage transition states, generating a structure–activity map that prioritises catalysts for wet-lab testing — extending Byron/Sadow/Delferro 2026 methodology to elastane.

Strictly still requires wet lab:

- Final activity validation (any computational kcat/Km is unreliable to >1 order of magnitude).
- Real-fibre substrate behaviour (crystallinity, dye adsorption, additive interference).
- Scale-up under industrially relevant flow and solvent recovery.
- Toxicology (MDA release) of any depolymerisation pathway.

Where the ceiling is highest and most credible: the **de novo Ser-Sercis-Lys urethanase theozyme**, because (a) every required generative tool is open-source and freely available, (b) the GRASE result demonstrates that even mined natural sequences can hit kilogram-scale industrial activity, (c) the carbamate cleavage chemistry now sits in a designable window after the November 2025 Bauer/Baker amide-bond demonstration, and (d) the corresponding wet-lab proof would constitute a clear "AlphaFold moment" for polymer-deconstruction enzymes.

PART D — The Precise Unsolved Seam

Synthesising across A–C, the elastane × AI-molecular-design landscape contains **three concentric layers of "openness"**:

D1. Genuinely scientifically unsolved (no demonstrated solution anywhere)

- **De novo computational design of a urethanase / carbamate hydrolase** with industrially relevant activity on aryl-carbamate model substrates or on elastane fibre itself. **No paper exists.** The Ser-Sercis-Lys triad of the amidase-signature superfamily has never been recreated in a designed scaffold; the Baker lab's de novo metalloprotease (Nov 2025) and de novo serine hydrolases (May 2025) are the nearest non-natural designs but neither targets aryl carbamates. Concrete sub-problems:
    - Specifying the carbamate transition state and oxyanion-hole geometry in a theozyme grammar that RFdiffusion2/3 can consume.
    - Achieving aromatic-tolerant S1 pocket geometry for MDA/MDI segments.
    - Marrying urethanase active sites with **DEG/alcohol-co-solvent tolerance** (the GRASE breakthrough condition) in a designed enzyme.
    - Designing a *fibre-active* variant — i.e., productive binding to crystalline polymer surfaces, addressed for PETases by Tournier and Alper's groups but never for urethanases.
- **De novo organocatalyst or metal-catalyst design specifically for selective urethane cleavage** (preserving polyester and polyamide bonds in the same blend). The Byron/Sadow/Delferro ML-bimetallic methodology has never been applied to PU/elastane.
- **A predictive ML model that recommends solvent/catalyst combinations outperforming DMSO+DBN at lower temperature / lower DBN loading**, validated experimentally. COSMO-RS-on-elastane oligomers has not been published.

D2. Solved in the lab, unsolved at scale

- **AbPURase chemo-enzymatic glycolysis** (Chen *Science* 2025): solved at kilogram on PU foam, **not transferred to elastane fibre**. The fibre's crystallinity, dye loading, and core-spun yarn geometry have not been demonstrated to be enzyme-accessible.
- **UMG-SP2 engineering** (Li *Adv. Sci.* 2025): 30.7-fold improvement in lab, not transferred to elastane fibre or industrial DEG conditions.
- **DMSO+DBN selective elastane removal** (Ferreira 2025): solved at lab-scale 30–60 min batch; not yet a continuous pre-treatment integrated into a sortation/recycling line.
- **Chitosan recoverable coating** (Chazot/Northwestern 2025): proof-of-concept on filaments; not deployed at mill scale; not shown to survive dye / finish processing.

- **Hyperspectral elastane detection** (Refiberd, Specim, PICVISA): solved for >2% bulk composition; **trace-level (<1%) elastane detection on dark/black garments and through complex finishes remains the open challenge** (and is publicly stated as such by Specim/PICVISA).

D3. Pure-software contributions that would be novel and credible

This is the highest-leverage layer for a credible high-ceiling deliverable:

- **Open-source "ElastaneDesigner" theozyme + scaffold pipeline**: an RFdiffusion3 → LigandMPNN → PLACER → AlphaFold3 self-consistency loop targeting the Ser-Sercis-Lys urethanase active site with an MDI-segment-binding S1 pocket, releasing the designed protein backbones, sequences, and computational scores as the world's first published de novo urethanase candidates. (No wet-lab confirmation required for novelty; the *publication of designed sequences* alone would be a first.)
- **Open ML-COSMO-RS solvent/catalyst recommender** for selective elastane cleavage that improves on DMSO+DBN, with predictions ranked by selectivity, toxicity (REACH/CLP flags), and energy cost. Deployable as a webapp.
- **A federated, open-licensed hyperspectral elastane-blend dataset and benchmark**, releasing labelled NIR/SWIR spectra of children's apparel blends with trace elastane percentages. (Refiberd, Specim, PICVISA have called out trace-elastane detection as the open problem; no public benchmark exists.)
- **An ML "stream router" classifier** that takes hyperspectral spectra + garment metadata and recommends the optimal chemistry path (DMSO+DBN selective dissolution → PET hydrolysis vs. ZnO glycolysis vs. enzymatic vs. mechanical), with uncertainty quantification — i.e., the "MutCompute for textile streams."
- **Mechanism-conditioned MLIP fine-tuning** (MACE-MP-0 + OC20 transfer) on aryl-carbamate cleavage transition states across Ru, Pt, Mo, Mn, Zn bimetallics; produce a candidate-catalyst ranking compatible with the Byron/Sadow/Delferro methodology but for PU/elastane.

The combination — **GRASE has proved nature-mined urethanases work; FAST-PETase has proved ML-engineered hydrolases work; Bauer/Baker (Nov 2025) has proved de novo amide-bond cleavage works; no one has yet closed the loop to a de novo Ser-Sercis-Lys urethanase for elastane** — is the precise, defensible, "this shouldn't be possible yet" flag.

Recommendations (for Phase 1 idea-generation aiming)

1. **Aim idea generation at the de novo urethanase theozyme + scaffold problem.** Specifically, an open-source pipeline that emits, scores, and visualises candidate de novo enzymes for the Ser-Sercis-Lys carbamate-cleavage transition state in a designed fold,

conditioned on MDA/MDI-binding S1 pocket. Benchmark thresholds: any single design with self-consistency RMSD <2 Å, AF3 pLDDT >85 across the active site, PLACER ensemble preorganisation >50%, and a credible carbamate transition-state geometry would be a publishable first.

2. **Stage 2 fallback: ML-COSMO-RS solvent recommender beating DMSO+DBN.** Benchmark thresholds: a predicted solvent/catalyst pair with COSMO-RS-predicted χ_elastane < χ_DMSO+DBN at lower predicted temperature (⩽80 °C) AND lower predicted toxicity (preferably non-REACH-Annex-XVII), validated against literature mass-loss data where overlap exists.
3. **Stage 3 fallback: open hyperspectral trace-elastane benchmark + AutoML classifier.** Benchmark thresholds: ROC-AUC > 0.9 for detecting <1% elastane on dark garments using ⩽20 ms inference, on a publicly released dataset.
4. **Decision rule**: If by hour 24 the de novo enzyme pipeline has produced fewer than 100 distinct designs passing AF3 self-consistency, fall back to Stage 2; if Stage 2 cannot benchmark within 24 hours, fall back to Stage 3.
5. **Always include the children's-apparel dermal-safety overlay** in the framing: any recovered material destined for baby/infant textiles must demonstrably exclude MDA contamination, which makes the *selectivity* (not just activity) of the chosen chemistry the binding constraint — and selectivity is exactly what generative protein design and ML solvent screening can optimise that brute-force chemistry cannot.

Caveats and epistemic notes

- The "no-evidence-of-existence" claim for de novo urethanase design is necessarily incomplete: confidential industrial work at Samsara Eco, Carbios, Protein Evolution Inc., Liberation Labs, Profluent, Cradle.bio, Epoch Biodesign, or others could exist without public disclosure. Samsara's "AI-designed enzymes" language is marketing-grade and does not specifically claim *de novo* generative design — published/patented sequences appear to be ML-evolved variants of natural cutinases/amidases.
- Bauer/Baker (Nov 2025 bioRxiv) and Ahern et al. RFdiffusion2 (*Nat. Methods* Jan 2026) are recent enough that follow-ups specifically targeting urethanes may emerge before this hackathon concludes; the window is genuinely narrow.
- The Ferreira et al. DMSO+DBN paper (*Polymers* 2025) is the most current selective-elastane-destruction benchmark; older work (Boschmeier, Phan, De Smet, De Clerck Ghent/Centexbel) is the prior art they cite and build on.
- All quoted experimental percentages (98.6% PU degradation, 38× PETase activity, 30.7× UMG-SP2 activity, 36% one-shot hit rate / >10^8-fold acceleration for de novo metalloprotease, 380,000 stable GNoME crystals out of 2.2M predicted, 81–100%

DMSO+DBN elastane mass loss, 35% Roica V550 degradation in 270 days, etc.) are quoted verbatim from the indicated primary literature or company sources.

- Throughput claims from startups (Samsara's "20 minutes", Refiberd's "7 million pounds per camera", Resortecs's "90–95% recyclability") should be treated as press-release-grade rather than independently verified industrial figures.
- Lululemon × Samsara is a *conditional offtake* ("could see"), not a guaranteed 20% portfolio share, and targets nylon 6,6 / polyester — not elastane.
- The Carbios funding figure of €250+ million total equity raised (per BNP Paribas / Carbios FY2023 results, GlobeNewswire April 2024, with the July 2023 round being €141 million gross) supersedes earlier press-cycle figures.
- All preprints cited from late 2025 / early 2026 (Bauer et al. metalloprotease, RFdiffusion3 Butcher/Krishna/Baker, Proteus2 metalloprotease for Aβ) are not yet peer-reviewed at the time of this writing.